

Brindlewythe Biosciences Ltd, a Tarnovell company

AP02 second-generation safety panel with CYP3A4 concentration response

Final Report

Study BW21_0453 (LIMS B-ACC-2021-07710)

Client: Avelnelwick Pharmaceuticals Ltd

Study Director: Dr L. Benmezroun

Report date: 25 November 2021

1 Summary

42 compounds from the Avelnelwick aminopyrimidine regioisomer series 02 were tested for Avelnelwick Pharmaceuticals Ltd in the hERG automated patch clamp, 5-isoform CYP panel, 3A4 concentration response. Testing started on 08 November 2021 and results were reported on 25 November 2021. All runs met plate acceptance criteria. No compound inhibited hERG with an IC₅₀ below 1 µM. Most compounds were weak or inactive (IC₅₀ at or above 10 µM, or above the top concentration). Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2 was rarely affected.

Table 1 Study summary

Metric	Value
Project	BW21_0453
Client	Avelnelwick Pharmaceuticals Ltd
Compounds tested	42
Series / chemotype	Avelnelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM, CYP3A4 inhibition IC ₅₀
Turnaround (calendar days)	17
Price	GBP 15,900
hERG IC ₅₀ <1 µM	0
hERG IC ₅₀ 1-10 µM	20
hERG IC ₅₀ ≥10 µM	22
hERG median IC ₅₀ (µM)	10
E-4031 control IC ₅₀ (nM)	14
CYP1A2 >50% inhibition @10 µM	1
CYP2C9 >50% inhibition @10 µM	3
CYP2C19 >50% inhibition @10 µM	11
CYP2D6 >50% inhibition @10 µM	7
CYP3A4 >50% inhibition @10 µM	13
Compounds with any CYP >50%	17
Ketoconazole (3A4) IC ₅₀ (µM)	0.029

2 Objective

To rank the screening set for hERG automated patch clamp, 5-isoform CYP panel, 3A4 concentration response liability so that Avernell Pharmaceuticals Ltd could choose analogues for progression.

3 Test items

42 compounds were received as 10 mM DMSO solutions in one test lot on 08 November 2021 and logged against B-ACC-2021-07710. Identity and concentration were taken as supplied by the client.

4 Methods

hERG automated patch clamp

HEK293 cells stably expressing hERG (Kv11.1) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells were held at -80 mV, depolarised to +40 mV for 2 s and repolarised to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final), with 3-5 minutes per concentration.

- Plate acceptance: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current amplitude ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- Positive control E-4031 on every run; IC₅₀ must fall within 5-50 nM
- IC₅₀ from a four-parameter logistic fit of mean tail-current inhibition; $n \geq 2$ cells per compound

5-isoform CYP panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m , in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation was measured by LC-MS/MS against vehicle (0.1% DMSO) incubations.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Results are reported as mean percent inhibition relative to vehicle. Values above 50% at 10 μ M are flagged for follow-up; small negative values indicate no inhibition.

3A4 concentration response

Compounds were run in a 7-point concentration response (top concentration 50 μ M, 3-fold dilutions) against CYP3A4 with midazolam or testosterone as probe; ketoconazole was the positive control.

5 Results

hERG results

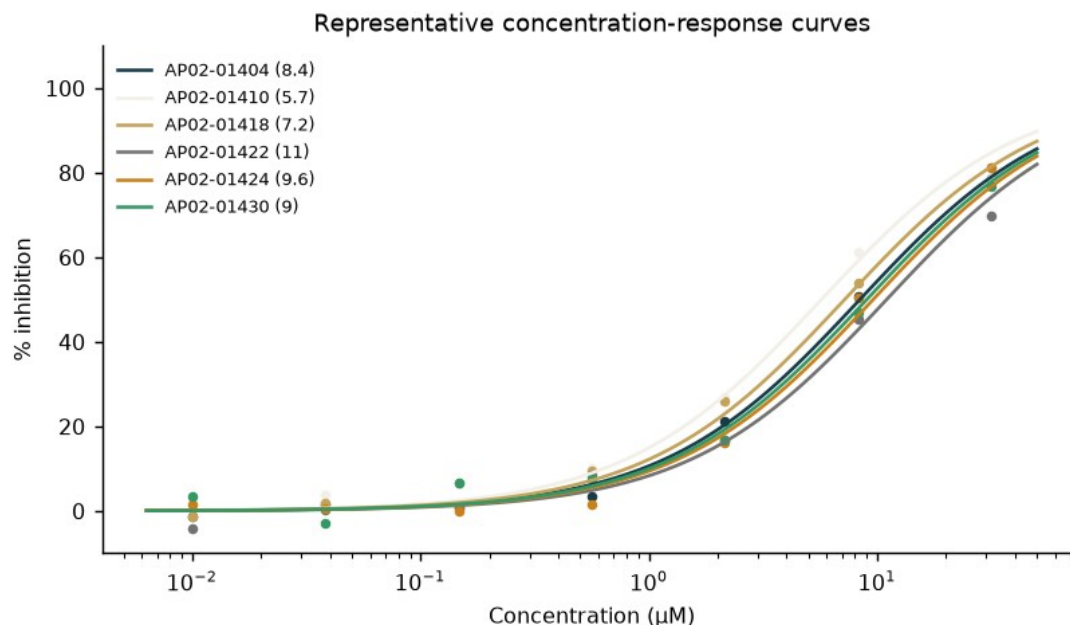
All 42 compounds were tested. No compound inhibited hERG with an IC₅₀ below 1 μ M. Most compounds were weak or inactive (IC₅₀ at or above 10 μ M, or above the top concentration). The median IC₅₀ was 10 μ M (0 below 1 μ M, 20 between 1 and 10 μ M, 22 at or above 10 μ M). E-4031 gave an IC₅₀ of 14 nM, within the acceptance range.

Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ rep 1 (μ M)	IC ₅₀ rep 2 (μ M)	IC ₅₀ rep 3 (μ M)	IC ₅₀ mean (μ M)	% inhibition @10 μ M	n	QC note
AP02-01400	>30	>30		>30	23	2	
AP02-01404	8.4	19		13	47	2	
AP02-01410	5.7	7.2		6.4	56	2	
AP02-01416	>30	21		>30	22	2	
AP02-01418	7.2	11	12	9.8	48	3	
AP02-01422	11	15		13	42	2	
AP02-01424	9.6	19	11	13	46	3	
AP02-01430	9.0	12		10	59	2	
AP02-01434	20	15		18	37	2	
AP02-01442	24	21		23	30	2	
AP02-01446	11	8.9		9.9	53	2	
AP02-01448	4.0	4.3		4.1	69	2	
AP02-01454	3.7	5.4		4.5	71	2	
AP02-01458	6.9	5.0		5.8	64	2	
AP02-01464	10	8.5	8.0	8.9	50	3	
AP02-01472	21	14	15	16	42	3	
AP02-01476	11	6.6		8.4	61	2	
AP02-01484	18	19		18	35	2	
AP02-01488	4.6	4.2		4.4	75	2	
AP02-01494	3.1	3.9		3.5	81	2	
AP02-01496	14	26		19	40	2	
AP02-01500	5.8	8.7		7.1	63	2	
AP02-01506	28	>30	>30	>30	26	3	
AP02-01508	12	11	13	12	47	3	
AP02-01514	12	17		14	37	2	
AP02-01518	2.8	2.2		2.5	84	2	
AP02-01524	29	22		25	28	2	
AP02-01526	29	18		23	29	2	
AP02-01532	4.7	3.4		4.0	68	2	
AP02-01536	3.5	6.2		4.7	71	2	
AP02-01538	17	19		18	34	2	
AP02-01544	16	24		20	32	2	
AP02-01550	11	9.6		10	46	2	
AP02-01556	9.7	9.5		9.6	49	2	
AP02-01560	6.7	11		8.7	57	2	
AP02-01568	10	7.8		8.9	57	2	
AP02-01572	7.6	9.3		8.4	48	2	
AP02-01578	13	12	10	12	46	3	
AP02-01580	5.5	6.5		5.9	62	2	
AP02-01584	14	17	12	14	42	3	
AP02-01586	11	8.6	7.1	8.8	55	3	Run 2 rejected (E-4031 outside

							range); repeated
AP02-01590	15	13		14	41	2	

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC₅₀ 14 nM (acceptance 5-50 nM)



CYP inhibition results

Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. In total 17 compounds inhibited at least one isoform by more than 50% at 10 μ M (CYP3A4: 13; CYP2D6: 7; CYP2C19: 11; CYP2C9: 3; CYP1A2: 1). Ketoconazole inhibited CYP3A4 with an IC₅₀ of 0.029 μ M.

Table 3 Percent inhibition at 10 μ M

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
AP02-01400	21	13	45	14	65
AP02-01404	-2	-6	2	-8	9
AP02-01410	11	36	85	99	84
AP02-01416	6	-4	3	-7	-2
AP02-01418	2	0	15	5	3
AP02-01422	-1	6	0	-4	5
AP02-01424	-0	3	2	0	3
AP02-01430	21	5	39	13	49
AP02-01434	9	10	26	10	85
AP02-01442	-3	32	65	-2	24
AP02-01446	16	82	83	14	91
AP02-01448	1	4	-0	13	67
AP02-01454	1	9	30	5	21
AP02-01458	14	35	67	45	22
AP02-01464	-5	-1	0	-2	-0
AP02-01472	-1	5	3	16	40
AP02-01476	2	23	52	58	24
AP02-01484	-5	8	23	16	1
AP02-01488	3	1	25	4	35
AP02-01494	64	11	62	16	62
AP02-01496	-4	-2	-6	5	-1
AP02-01500	1	11	0	12	3
AP02-01506	7	0	-8	3	-0

AP02-01508	6	37	11	94	67
AP02-01514	-0	-3	44	2	-0
AP02-01518	5	4	92	100	66
AP02-01524	32	3	13	12	3
AP02-01526	-2	1	-1	7	6
AP02-01532	-0	12	-6	64	20
AP02-01536	-0	-1	78	68	55
AP02-01538	3	-1	3	35	-5
AP02-01544	43	6	64	46	68
AP02-01550	7	3	-2	8	-4
AP02-01556	4	70	89	39	70
AP02-01560	-0	2	-3	-0	-4
AP02-01568	-4	11	12	28	33
AP02-01572	0	5	2	18	11
AP02-01578	48	10	45	33	89
AP02-01580	6	53	76	53	67
AP02-01584	14	-1	-3	14	26
AP02-01586	-7	-1	24	6	3
AP02-01590	0	6	39	14	35

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.9; sulfaphenazole (CYP2C9) 0.33; ticlopidine (CYP2C19) 1.6; quinidine (CYP2D6) 0.055; ketoconazole (CYP3A4) 0.029

CYP3A4 concentration response

Table 4 CYP3A4 IC₅₀ (μ M)

Compound	IC ₅₀ (μ M)	Probe substrate
AP02-01400	7.9	midazolam
AP02-01404	>50	midazolam
AP02-01410	2.0	midazolam
AP02-01416	>50	midazolam
AP02-01418	>50	midazolam
AP02-01422	>50	testosterone
AP02-01424	>50	midazolam
AP02-01430	11	midazolam
AP02-01434	1.6	testosterone
AP02-01442	28	midazolam
AP02-01446	0.83	midazolam
AP02-01448	2.7	midazolam
AP02-01454	>50	midazolam
AP02-01458	>50	midazolam
AP02-01464	>50	midazolam
AP02-01472	14	testosterone
AP02-01476	34	midazolam
AP02-01484	>50	testosterone
AP02-01488	41	midazolam
AP02-01494	4.3	testosterone
AP02-01496	>50	testosterone
AP02-01500	>50	midazolam
AP02-01506	>50	midazolam
AP02-01508	4.9	midazolam
AP02-01514	>50	testosterone
AP02-01518	4.6	testosterone
AP02-01524	>50	testosterone
AP02-01526	>50	midazolam
AP02-01532	31	midazolam
AP02-01536	11	testosterone
AP02-01538	>50	midazolam
AP02-01544	7.7	testosterone
AP02-01550	>50	midazolam

AP02-01556	3.5	testosterone
AP02-01560	>50	midazolam
AP02-01568	36	midazolam
AP02-01572	42	midazolam
AP02-01578	1.4	midazolam
AP02-01580	6.7	midazolam
AP02-01584	22	testosterone
AP02-01586	>50	testosterone
AP02-01590	31	midazolam

7-point curve, top 50 μ M; ketoconazole control

6 Discussion

No compound inhibited hERG with an IC₅₀ below 1 μ M. Most compounds were weak or inactive (IC₅₀ at or above 10 μ M, or above the top concentration). Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. These are screening data; binding-site and mechanism statements are outside the scope of this study.

7 Conclusion

The data support ranking within the set. Recommended next step: progress the compounds with the widest hERG margin and no CYP flag, and confirm any CYP3A4 flags by concentration response.

Name	Role	Signature	Date
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Ines Alvarado	Assay Quality Coordinator	<i>Alvarado</i>	24 November 2021